Linear Algebra and Its Applications

Section 2.3: Characterizations of Invertible Matrices

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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2.3 Characterizations of invertible matrices

The Invertible Matrix Theorem (IMT) is a collection of equivalent tests for an $n \times n$ matrix. Invertibility, n pivots, independent columns, columns spanning $\mathbb{R}^n$, a trivial null space, and a one-to-one and onto map $x \mapsto Ax$ all describe the same situation. The word *square* is essential: a rectangular matrix can have independent columns without spanning its whole codomain.

For implications, distinguish a proof from a plausible example. A true implication needs an argument for *every* allowed matrix; a false implication needs just one counterexample with true premise and false conclusion. The zero matrix and identity matrix are useful tests, but their special properties should not be mistaken for universal properties.

Practice problem 2.3.P1: Repeated columns prevent invertibility

The three columns of the supplied matrix are $(2, 2, 2)^T$, $(3, 3, 3)^T$, and $(4, 4, 4)^T$. In particular column 2 is $3/2$ times column 1, so the columns are dependent. The IMT says an invertible square matrix has independent columns. Therefore this matrix is not invertible.

Practice problem 2.3.P2: Negate an existence statement correctly

Statement (g) says: for *every* $b \in \mathbb{R}^n$ there exists a solution to $Ax = b$. Its negation is: there exists *at least one* b for which $Ax = b$ is inconsistent. It does not say every system is inconsistent; indeed $Ax = 0$ always has $x = 0$ as a solution. The IMT also implies that this A is not invertible.

Practice problem 2.3.P3: Recognize a noninvertible product

The matrix AB is square. A nontrivial solution of $(AB)x = 0$ violates the trivial-null-space condition in the IMT, so AB is not invertible. We cannot conclude that *both* A and B are singular; one singular factor suffices to make a product singular. For instance $A = 0$, $B = I$ has exactly one singular factor.

Exercise 2.3.1: A two-by-two determinant test

For $A = \begin{bmatrix} 5 & 7 \\ -3 & -6 \end{bmatrix}$, $ad - bc = -30 + 21 = -9 \neq 0$. The two-by-two inverse criterion from Section 2.2 therefore gives **invertible**. No complete inverse calculation is needed.

Exercise 2.3.2: A multiple row

The second row $(6, -3)$ is $-3/2$ times the first row $(-4, 2)$. Eliminating it leaves only one pivot, fewer than two. Thus the matrix is **not invertible**. Equivalently the determinant is $12 - 12 = 0$.

Exercise 2.3.3: A triangular matrix with nonzero diagonal

The matrix is lower triangular with diagonal $3, -4, -3$. The first pivot 3 removes the entries below it; the second pivot -4 removes the next entry below it; the last pivot remains -3. None vanishes.

There is a pivot in every row and column, so the IMT gives **invertible**. The triangular structure saves a full reduction.

Exercise 2.3.4: A zero row

The second row is all zeros. Thus there can be at most two pivot rows in this 3×3 matrix. Since invertibility would require three pivots, the matrix is **not invertible**.

Exercise 2.3.5: A zero column

Column 2 is zero, so $Ae_2 = 0$ with $e_2 \neq 0$. This is a nontrivial null vector. The IMT therefore gives **not invertible**; equivalently the columns are dependent because one of them is the zero vector.

Exercise 2.3.6: A short elimination gives the last pivot

Use $R_3 \leftarrow R_3 + 3R_1$ to obtain the last row $(0, -3, -18)$, then $R_3 \leftarrow R_3 + (3/4)R_2$. The resulting echelon matrix is

$$\begin{bmatrix} 1 & -3 & -6 \\ 0 & 4 & 3 \\ 0 & 0 & -63/4 \end{bmatrix}.$$

Its three diagonal entries are nonzero pivots, so the original matrix is **invertible**. The operations preserve row equivalence.

Exercise 2.3.7: Four pivots from a few row replacements

Apply $R_2 \leftarrow R_2 + 3R_1$, $R_3 \leftarrow R_3 - 2R_1$, and then $R_4 \leftarrow R_4 - (1/4)R_2$. This gives

$$\begin{bmatrix} -1 & -3 & 0 & 1 \\ 0 & -4 & 8 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Every diagonal position is a pivot. Hence the original 4×4 matrix is **invertible**. There is no need to clear the entries above the pivots.

Exercise 2.3.8: Read pivots from an upper triangular matrix

The given matrix is already in echelon form with nonzero diagonal entries $3, 1, 2, 1$. It has four pivots, so it is **invertible** by the IMT. The entries above the diagonal do not change this conclusion.

Exercise 2.3.9: A computer-assisted exact pivot test

Using rational row reduction on the supplied integer matrix gives an echelon matrix

$$\begin{bmatrix} 4 & 0 & -3 & -7 \\ 0 & 9 & 9/2 & -3/2 \\ 0 & 0 & 71/4 & 365/12 \\ 0 & 0 & 0 & -1336/213 \end{bmatrix}.$$

For reproducibility, elimination here uses $R_i \leftarrow R_i - (a_{ik}/a_{kk})R_k$ below each successive pivot. All four pivots are nonzero; therefore the matrix is **invertible**. An exact program check is `Matrix(A).rref()`, using SymPy, which returns I_4 and pivot columns $(0, 1, 2, 3)$ (zero-based indices). These exact calculations avoid confusing a very small pivot with zero.

Exercise 2.3.10: Five pivots despite a small determinant

For the supplied 5×5 integer matrix, exact elimination, including one row interchange when a candidate pivot is zero, gives

$$\begin{bmatrix} 5 & 3 & 1 & 7 & 9 \\ 0 & 2/5 & 4/5 & -2/5 & -94/5 \\ 0 & 0 & 1 & -21 & 7 \\ 0 & 0 & 0 & 1 & 34 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}.$$

There are five nonzero pivots, hence the matrix is **invertible**. Its inverse can nevertheless be numerically sensitive; invertibility is an exact yes/no statement, while conditioning measures sensitivity. Exercise 43 quantifies that distinction. Exact SymPy row reduction returns I_5 with all five pivot columns.

Exercise 2.3.11: Five implications, with the square hypothesis retained

(a) **True.** A trivial null space means no free variables, so all n columns have pivots. For a square matrix that forces the reduced form to be I_n; this is an IMT equivalence.

(b) **True.** Spanning $\mathbb{R}^n$ forces n pivot rows. In an $n \times n$ matrix these are also n pivot columns, giving independence.

(c) **False.** Merely being $n \times n$ does not guarantee consistency for every right side. Take $A = 0$ and $b = e_1$; the equation $0x = e_1$ has no solution.

(d) **True.** A nontrivial homogeneous solution requires a free variable, so some column is not a pivot column. Thus there are fewer than n pivots.

(e) **True.** If A were invertible, then $(A^{-1})^T$ would be an inverse of A^T. The given noninvertibility of A^T excludes that possibility, so A is not invertible.

Exercise 2.3.12: Pay attention to onto versus into

(a) **True.** $AD = I$ is the right-inverse condition in the IMT, so A is invertible. Left multiplication by A^{-1} gives $D = A^{-1}$, and therefore $DA = I$.

(b) **False.** Every $n \times n$ matrix maps $\mathbb{R}^n$ *into* $\mathbb{R}^n$, including $A = 0$, whose reduced form is not I. Mapping *onto* would be a stronger premise and would imply the conclusion.

(c) **True.** For a square matrix, independent columns and columns spanning $\mathbb{R}^n$ are equivalent by the IMT.

(d) **False.** Consistency for every b implies invertibility, hence the transformation *is* one-to-one. The identity matrix is a concrete counterexample to the claimed negative conclusion.

(e) **False.** Consistency for just one right side says nothing about uniqueness. For $A = 0$, $b = 0$, every vector is a solution.

Exercise 2.3.13: When an upper triangular matrix is invertible

It is invertible *if and only if every diagonal entry is nonzero.* If all are nonzero, eliminate above the diagonal from the last row upward, then scale the rows; this reduces the matrix to I_n. For the converse, suppose diagonal entry k is zero. In an upper triangular matrix, each of the first $k-1$ columns has zero entries from row k downward. Column k also has zero entries from row k downward: its diagonal is zero, and its lower entries vanish by triangularity. Thus the first k columns live entirely in their first $k-1$ coordinates. A homogeneous system with k variables but only $k-1$ potentially nonzero equations has a free variable, so these k columns are dependent. Their dependence extends to all columns by assigning zero coefficients to the remaining columns. The IMT therefore excludes invertibility.

Exercise 2.3.14: The lower triangular case

Again the criterion is that every diagonal entry be nonzero. One way to avoid repeating the argument is to transpose: A is lower triangular exactly when A^T is upper triangular, with the same diagonal. The IMT says A is invertible exactly when A^T is invertible. Applying Exercise 13 to A^T proves both directions of the claimed criterion.

Exercise 2.3.15: Nonspanning columns exclude an inverse

No. The IMT says an invertible 4×4 matrix has columns spanning $\mathbb{R}^4$. The proposed failure of spanning is the negation of one of its equivalent conditions, so it rules out invertibility.

Exercise 2.3.16: Independence of the transposed columns

If A is invertible, then A^T is invertible, with inverse $(A^{-1})^T$. Applying the independent-column condition of the IMT to A^T shows that its columns are independent. Equivalently the rows of A are independent.

Exercise 2.3.17: Identical columns produce an explicit dependence

No. If columns i, j are equal with $i \neq j$, then $A(e_i - e_j) = a_i - a_j = 0$. The vector $e_i - e_j$ is nonzero, so the null space is nontrivial and A cannot be invertible.

Exercise 2.3.18: Identical rows also rule out invertibility

No. Equal rows of A become equal columns of A^T. Exercise 17 makes A^T singular, and the transpose equivalence in the IMT then makes A singular too.

Exercise 2.3.19: Seven independent columns give every right side exactly once

Since D is 7×7 and has independent columns, the IMT makes it invertible. For *every* $b \in \mathbb{R}^7$, the system $Dx = b$ therefore has the unique solution $x = D^{-1}b$. Both existence and uniqueness are part of the conclusion; neither needs a special choice of b.

Exercise 2.3.20: Universal consistency for a square system implies uniqueness

It is not possible to have more than one solution for any b. Universal consistency gives a pivot in every one of the five rows. Because there are five columns too, all five variables are pivot variables, leaving no free variable. The matrix is invertible, and every system has exactly one solution.

Exercise 2.3.21: Multiple solutions exclude spanning

If $Cu = v$ has two different solutions u_1, u_2, then $C(u_1 - u_2) = v - v = 0$ with $u_1 - u_2 \neq 0$. Thus C is singular by the IMT. Its columns therefore cannot span $\mathbb{R}^n$. Notice that the original right side v is still attained; the failure concerns some other right sides.

Exercise 2.3.22: A one-sided square inverse forces commutation

From $EF = I$, the IMT makes E invertible. Multiplying on the left by E^{-1} gives $F = E^{-1}$, so $FE = I$ as well. Hence $EF = FE$, which is exactly the definition of commuting matrices. This conclusion uses squareness; rectangular one-sided inverses need not reverse.

Exercise 2.3.23: A missed output implies a nontrivial null vector

If $Fx = y$ is inconsistent for some y, then F is not onto. By the IMT this square matrix is singular, so $Fx = 0$ has a nontrivial solution. In fact if $z \neq 0$ is such a solution, every tz, $t \in \mathbb{R}$, is a solution; thus the homogeneous system has infinitely many solutions.

Exercise 2.3.24: A failure to reduce to the identity

If G cannot be reduced to I_n, the IMT makes it singular. Consequently its columns are dependent and do not span $\mathbb{R}^n$. There are fewer than n pivots. These are equivalent descriptions of the same failure, not alternative cases to choose between.

Exercise 2.3.25: Verify the boxed fact about a one-sided inverse

Assume square matrices A, B satisfy $AB = I$. Then A has a right inverse, so the IMT implies that A is invertible. The equation gives $B = A^{-1}AB = A^{-1}$. Therefore B is invertible too, with inverse A,

and $BA = I$. This proves every assertion in the boxed statement preceding Example 1, rather than merely assuming a right inverse is automatically two-sided.

Exercise 2.3.26: Independent columns lead to a spanning square

Independence of the columns of square A makes A invertible. The product $A^2 = AA$ is then invertible, with inverse $A^{-1}A^{-1}$. Applying the spanning condition of the IMT to A^2 proves that its columns span $\mathbb{R}^n$. Equivalently, A maps onto $\mathbb{R}^n$ and doing so twice still reaches every vector.

Exercise 2.3.27: An invertible product forces the first factor invertible

Let $W = (AB)^{-1}$. Then $A(BW) = (AB)W = I$. Thus A has a square right inverse BW, and the IMT makes A invertible. This proof does *not* apply the inverse-of-a-product formula to unverified inverses of A, B; their existence is what must be proved.

Exercise 2.3.28: An invertible product forces the second factor invertible

Suppose $Bx = 0$. Multiplying by A gives $(AB)x = 0$. Invertibility of AB implies $x = 0$. Hence B has trivial null space, and since B is square the IMT proves it is invertible. Alternatively, $(WA)B = I$ with $W = (AB)^{-1}$ gives a left-inverse test.

Exercise 2.3.29: One-to-one endomorphisms are also onto

The one-to-one condition says $Ax = 0$ has only $x = 0$, or equivalently the columns are independent. For $n \times n$ matrices the IMT makes A invertible and hence onto $\mathbb{R}^n$. Thus the transformation is a bijection with inverse $x \mapsto A^{-1}x$. The equal dimensions are crucial.

Exercise 2.3.30: Failure of injectivity also forces failure of surjectivity

Two solutions of $Ax = b$ give a nonzero homogeneous solution by subtraction. Therefore A is singular. By the IMT the associated transformation is not onto $\mathbb{R}^n$ either: at least one right side has no preimage. It can still attain the particular b mentioned in the premise.

Exercise 2.3.31: Prove uniqueness directly from pivot counting

What we must show. Use row reduction, without Theorems 5 or 8. Consistency for all b forces a pivot in every row; otherwise an appropriate right side creates an inconsistent zero-coefficient row. The square matrix has n rows and n columns, so these n pivots use all n columns. There are no free variables. Thus, for any fixed b, back-substitution determines each variable uniquely. The hypothesis gives existence, and this pivot argument gives uniqueness, as requested.

Exercise 2.3.32: Prove existence directly from a trivial null space

What we must show. Avoid invoking the IMT. If a column were not a pivot column, its variable could be chosen freely in the homogeneous system, producing a nonzero solution. Thus all n columns

have pivots. The n pivots occupy different rows, and there are only n rows, so every row has a pivot. Hence for every augmented right side the system is consistent: no zero coefficient row can create a contradiction. This gives a solution of $Ax = b$ for every b directly.

Exercise 2.3.33: Invert a two-dimensional linear transformation

The standard matrix is $A = \begin{bmatrix} -5 & 9 \\ 4 & -7 \end{bmatrix}$, whose determinant is $35 - 36 = -1$. Thus $A^{-1} = \begin{bmatrix} 7 & 9 \\ 4 & 5 \end{bmatrix}$. For an output (y_1, y_2), its unique input is

$$\boxed{T^{-1}(y_1, y_2) = (7y_1 + 9y_2,\ 4y_1 + 5y_2)}.$$

Multiplying either pair of matrices gives I_2, which verifies both compositions $T^{-1} \circ T$ and $T \circ T^{-1}$ are the identity.

Exercise 2.3.34: A second inverse transformation

Now $A = \begin{bmatrix} 2 & -8 \\ -2 & 7 \end{bmatrix}$ has determinant $14 - 16 = -2$, so $A^{-1} = \begin{bmatrix} -7/2 & -4 \\ -1 & -1 \end{bmatrix}$. Consequently

$$\boxed{T^{-1}(y_1, y_2) = (-\tfrac{7}{2}y_1 - 4y_2,\ -y_1 - y_2)}.$$

For example applying T to this pair gives first coordinate $-7y_1 - 8y_2 + 8y_1 + 8y_2 = y_1$ and second coordinate $7y_1 + 8y_2 - 7y_1 - 7y_2 = y_2$. The reverse composition follows from the same inverse matrix calculation.

Exercise 2.3.35: Use both inverse identities directly

Let $S = T^{-1}$, so $S(T(x)) = x$ and $T(S(y)) = y$. If $T(u) = T(v)$, apply S to get $u = v$; thus T is one-to-one. For every desired output y, the input $S(y)$ satisfies $T(S(y)) = y$; thus T is onto. As a second proof, Theorem 9 identifies invertibility of T with invertibility of its square standard matrix, and the IMT then gives both injectivity and surjectivity.

Exercise 2.3.36: Two inputs sharing an output rule out onto

The equality $T(u) = T(v)$ for $u \neq v$ means T is not one-to-one. Its square standard matrix therefore fails one IMT condition, so it also fails the onto condition. Hence T cannot map $\mathbb{R}^n$ onto $\mathbb{R}^n$. One can also subtract to see $T(u - v) = 0$ with $u - v \neq 0$.

Exercise 2.3.37: A one-sided linear inverse in equal dimensions

Let A, B be the standard matrices of T, U. The hypothesis $T(U(x)) = x$ for all x implies $ABx = x$ for all x. Testing the standard basis vectors gives $AB = I$. The boxed result proved in Exercise 25 yields $BA = I$. Thus $U(T(x)) = BAx = x$ for every x. The conclusion depends on both linearity and the common finite dimension.

Exercise 2.3.38: Why the inverse function is unique

Since T is invertible, it is onto. Given arbitrary $v \in \mathbb{R}^n$, choose x with $T(x) = v$. Then

$$S(v) = S(T(x)) = x = U(T(x)) = U(v).$$

Every v was allowed, so $S = U$ as functions. We did not need to assume that S, U were linear; the surjectivity of T ensured their arguments covered the entire domain.

Exercise 2.3.39: An onto linear map in equal dimensions has an onto inverse

By the IMT applied to its standard $n \times n$ matrix, an onto T is invertible. For any x, taking $y = T(x)$ gives $T^{-1}(y) = x$; therefore T^{-1} is onto. If $T^{-1}(y_1) = T^{-1}(y_2)$, apply T to obtain $y_1 = y_2$; therefore it is one-to-one as well. So both T and its inverse are bijections of $\mathbb{R}^n$.

Exercise 2.3.40: Prove linearity of the inverse without assuming it

Let $x = S(u)$ and $y = S(v)$. The identity $T \circ S = \mathrm{id}$ gives $T(x) = u$, $T(y) = v$. By linearity of T, $T(x + y) = u + v$. Applying S and using $S \circ T = \mathrm{id}$ yields

$$S(u + v) = S(T(x + y)) = x + y = S(u) + S(v).$$

Likewise $T(cx) = cT(x) = cu$ gives $S(cu) = S(T(cx)) = cx = cS(u)$. These are additivity and homogeneity, so S is linear. Both inverse identities were used explicitly.

Exercise 2.3.41: Tiny data perturbations, large solution changes

(a) Solve both systems exactly.

For either right side (b_1, b_2), the inverse of the coefficient matrix is

$$A^{-1} = \frac{1}{4.5(1.1) - 3.1(1.6)} \begin{bmatrix} 1.1 & -3.1 \\ -1.6 & 4.5 \end{bmatrix} = \begin{bmatrix} -110 & 310 \\ 160 & -450 \end{bmatrix},$$

because $4.95 - 4.96 = -.01$. For the unrounded right side,

$$x_1 = -110(19.249) + 310(6.843) = 3.94, \qquad x_2 = 160(19.249) - 450(6.843) = .49.$$

For the rounded right side, the same products give $(2.90, 2.00)^T$. All these decimals are exact terminating decimals, not numerical approximations.

(b) Specify the meaning of percentage error.

Coordinatewise relative percentage errors are

$$100\frac{|2.90 - 3.94|}{3.94} = \frac{5200}{197}\% \approx 26.40\%, \qquad 100\frac{|2.00 - .49|}{.49} = \frac{15100}{49}\% \approx 308.16\%.$$

If a single vector error is wanted instead, the Euclidean relative error is $100\|\widetilde{x} - x\|_2/\|x\|_2 \approx 46.18\%$. Stating the norm matters: these are different legitimate summaries.

(c) Compute and interpret conditioning.

Executing `np.linalg.cond(A)` in NumPy gives the spectral (2-norm) condition number $\kappa_2(A) \approx 3362.9997$. The 1-norm and infinity-norm condition numbers both equal 4636; software conventions therefore need to be identified. The large value explains why very small changes of the measured right side can cause large changes in the recovered solution.

Exercise 2.3.42: Measure roundoff for the matrix in Exercise 9

What we must show. Compare a known vector with a solution recovered from its computed output. The following is an executed NumPy experiment using binary64 arithmetic:

```
import numpy as np
A = np.array([[4,0,-3,-7],[-6,9,9,9],
              [7,-5,10,19],[-1,2,4,-1]],dtype=float)
x = np.random.default_rng(23042).random(4)
b = A@x
x1 = np.linalg.solve(A,b)
print(np.linalg.cond(A))
print(np.max(np.abs(x1-x))/np.max(np.abs(x)))
print(np.abs((x1-x)/x))
```

The condition number is 11.3525089, and the normwise relative error $\|x_1 - x\|_\infty / \|x\|_\infty$ is approximately $4.52 \cdot 10^{-16}$. The chosen vector, rounded here for readability, is

$$x \approx (.98344656, .12402037, .75512517, .48561218)^T.$$

The worst coordinatewise relative error is about $2.13 \cdot 10^{-15}$; one coordinate was recovered exactly in this run.

Binary64 has 53 binary significant bits, about 15–16 decimal significant digits. Thus the observed normwise agreement is roughly 15 digits, with roughly 14–15 in the most sensitive coordinate. The expected worst-case loss suggested by $\log_{10} \kappa_2(A)$ is about 1.06 digits. The measured error is an observation for this vector, not a promise that every possible right side has identical accuracy. Libraries and hardware may alter the last few floating-point bits.

Exercise 2.3.43: Repeat with a more sensitive matrix

Use the same procedure but with the matrix in Exercise 10 and the seed below:

```
A = np.array([[5,3,1,7,9],[6,4,2,8,-8],
              [7,5,3,10,9],[9,6,4,-9,-5],
              [8,5,2,11,4]],dtype=float)
x = np.random.default_rng(23043).random(5)
b = A@x
x1 = np.linalg.solve(A,b)
print(np.linalg.cond(A))
print(np.max(np.abs(x1-x))/np.max(np.abs(x)))
```

The executed condition number is $\kappa_2(A) \approx 68622.4548$; the normwise relative error is approximately $6.01 \cdot 10^{-12}$. Thus this run retains about 11 decimal digits normwise, losing about 4–5 of the approximately 15–16 available digits. The theoretical sensitivity indicator $\log_{10} \kappa_2(A) \approx 4.84$ is consistent with that behavior. The exact pivot calculation in Exercise 10 proved invertibility, but it did not guarantee numerical robustness.

Exercise 2.3.44: The last inverse column of a fifth-order Hilbert matrix

To obtain column 5 of H^{-1} without the other columns, solve $Hx = e_5$, where $H_{ij} = 1/(i + j - 1)$ for $1 \le i, j \le 5$. An exact rational solve gives

$$\boxed{x = (630, -12600, 56700, -88200, 44100)^T}.$$

It can be verified by multiplying the five rows of H by this column; the results are $0, 0, 0, 0, 1$ respectively. For example the first is $630 - 6300 + 18900 - 22050 + 8820 = 0$. A numerical version is

```
H = np.array([[1/(i+j+1) for j in range(5)]
              for i in range(5)])
x = np.linalg.solve(H,np.eye(5)[:,4])
print(np.linalg.cond(H))
```

Here $\kappa_2(H) \approx 476607.25 \approx 10^{5.68}$. Starting with about 15–16 decimal digits, the condition estimate allows a loss of about six digits, so about 9–10 digits is a conservative expectation. Our executed solve has largest relative coordinate error about $1.93 \cdot 10^{-12}$ against the exact column, roughly 11–12 digits of agreement. A bound on possible loss need not be reached for this particular right side.

Exercise 2.3.45: A twelfth-order Hilbert inverse can be numerically unreliable

Use a numerical inverse, rather than an exact rational one, for this experiment:

```
H = np.array([[1/(i+j+1) for j in range(12)]
              for i in range(12)])
B = np.linalg.inv(H)
R = H@B-np.eye(12)
print(np.linalg.cond(H))
print(np.max(np.abs(R)))
print(np.linalg.norm(R,np.inf))
```

In the executed NumPy binary64 run, the outputs are approximately $1.64 \cdot 10^{16}$, 0.0892, and 0.320, respectively. Thus the largest entry of the computed product's error is nearly 0.09: HB is visibly different from the identity. The computed B_{11} is approximately 142.655, whereas the exact Hilbert inverse has $(H^{-1})_{11} = 12^2 = 144$.

The exact rational Hilbert matrix is invertible, but its condition number is comparable to the reciprocal of binary64 rounding precision. Ordinary floating-point arithmetic therefore has essentially no reliable significant digits guaranteed in an inverse computation of this size. Different numerical libraries may produce somewhat different inaccurate answers; none of that contradicts exact invertibility. Using higher precision and checking against exact rational arithmetic are ways to investigate further, not reasons to silently report the floating-point product as I.